



Great Learning

6-8 minutes

1. Why do AI agents need external tools at all? Can't LLMs solve problems on their own?

LLMs are excellent at reasoning and language generation, but they are fundamentally limited in three ways:

- They cannot access live or private data
- They cannot execute actions (run code, query databases, call APIs)
- They do not verify outputs against real systems

External tools allow agents to:

- Compute exact answers (Python)
- Retrieve structured business data (SQL)
- Fetch up-to-date information (Search)
- Perform real-world actions (APIs)

Without tools, agents remain advisory. With tools, they become operational.

2. What types of tools do agents typically use, and why?

Agents can use many tool categories, each suited to different tasks:

- Search tools - Query search engines (Google, Bing, etc.) for current information. For example, a web search tool lets the agent find up-to-the-minute news or research.
- SQL/Database tools - Run database queries (SQL or API-based) against company data (customer records, inventory, etc.). This lets the agent retrieve or update structured business data directly.
- Python (code) tools - Execute code (often Python) to do precise computation or data processing. A code-interpreter tool can run scripts (using libraries like Pandas/NumPy) for analytics or even call other systems.
- Generic API tools - Call external services via HTTP (REST/SOAP APIs). For instance, an agent could use a generic API caller to update a CRM record, send an email (Gmail API), check the weather, or trigger an e-commerce order.

Each tool type gives different capabilities: search finds info; SQL accesses databases; code (Python) performs calculations; APIs enable integration with virtually any service.

3. How do agents decide which tool to use?

